HATELENS: Hateful Video Detection with Cues from Dual Perspectives

Anonymous ACL submission

Abstract

Hateful video detection poses greater challenges to the online ecosystem, as hateful content in video form spreads more rapidly and can have a stronger impact than traditional forms of dissemination. Existing detection methods fall short in understanding how hateful meaning is formed in videos and struggle to handle ambiguous cases involving implicit hate. In this work, we propose HATELENS, a dual-perspective paradigm that integrates cues from both internal and external environments. Internally, we leverage multimodal large language models (MLLMs) through a perception-to-cognition reasoning process, producing structured meaning formation for understanding hateful context in videos. Externally, we propose a test-time retrieval and adaptation mechanism that operates on fused multimodal representations to exploit annotated precedents, providing richer local evidence to help disambiguate implicit hate. Extensive experiments on three public datasets demonstrate that our method achieves state-of-the-art performance, with macro-precision improvements of up to 16%. Code of our project is available at...

Disclaimer: This paper contains content disturbing to some readers.

1 Introduction

The proliferation of short-form video platforms such as YouTube Shorts, TikTok, and Bilibili has created new vectors for the dissemination of hateful content (Das et al., 2023; Wang et al., 2024a). Unlike text-only hate speech, hateful videos embed harmful messages across multiple modalities (visual imagery, spoken language, on-screen text, and audio cues), making them particularly challenging to detect and moderate (Kiela et al., 2020; Lang et al., 2025). The multimodal nature of video hate demands methods that can reason across these diverse signals rather than merely aggregate them.

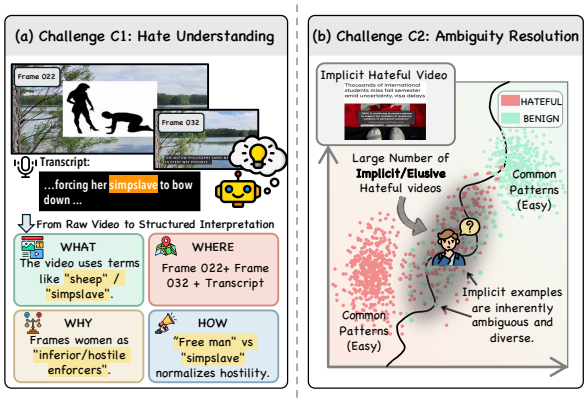

Figure 1: figure1 motivation.

Existing approaches to hateful video detection fall into two categories. *Multimodal feature fusion methods* extract representations from each modality using pre-trained encoders (e.g., BERT for text, ViT for frames, audio models for speech) and combine them through attention mechanisms (Das et al., 2023; Céspedes-Sarrias et al., 2025), mixture-of-experts architectures (Lang et al., 2025), or graph-based fusion (Yue et al., 2025). On the other hand, the rise of multimodal large language models (MLLMs) proves strong reasoning capability that can help both explainability and accuracy of detecting hate videos. Therefore, *MLLM-based methods* leverage MLLMs to analyze video content, typically through chain-of-thought (CoT) prompting that first describes visual and textual elements, then combines these descriptions into a judgment (Jing et al., 2025; Yang et al., 2025).

However, two important challenges remain underexplored. **(C1) Lack of structured meaning formation for hateful video understanding.** First, multimodal feature fusion methods model *what* is present in the video, but not *how* the content should be interpreted. This creates a dilemma in which the model learns surface-level features without learning to contextualize why the content is hateful,

thereby limiting the generalizability of the learned representations. Although the rise of MLLMs has created new opportunities to address this issue, current MLLM-based methods (Jing et al., 2025; Yang et al., 2025, 2026) still remain at the level of perceiving and analyzing surface features, or introduce prior bias by anchoring on a predefined judgment before reasoning. **(C2) Lack of effective ambiguity resolution for hateful video decisions.** Even when models learn to some extent *how* hate is expressed in video during training, the resulting representations are still likely to only support coarse-grained global decisions that mainly cover common patterns. However, hateful content is often expressed in implicit and diverse ways to evade detection, making such cases inherently ambiguous and placing them near the decision boundary. Although some prior work (Lang et al., 2025) has introduced external retrieval to alleviate this issue, these methods typically operate on early multimodal features with manually specified modality weights, making them less effective for handling the dynamic nature of the video ecosystem.

Brunswik's Lens Model implies that judgments are formed by integrating cues from both internal and external environments. Inspired by this, we propose HATELENS, which shifts the paradigm of hateful video detection from a direct perception to discriminative classification problem to a process of internal reasoning with structured meaning formation, followed by ambiguity resolution through external precedent case consultation. Specifically, the *P2C Generator* first leverages the MLLM to organize multimodal observations of a given video into cognitive evidence through perception-to-cognition deduction, producing a structured reasoning schema for meaning formation, including *what* the hateful content is, *where* it is located, *why* it should be classified as hateful, and *how* it may promote hateful activity, thereby deeply contextualizing the video. The *Schema-guided Modality Router* then jointly fuses all multimodal features with the above textual reasoning schema to produce more discriminative feature representations. Finally, to address ambiguity, the *Whitened Neighbor Interpolator*, a gradient-free module, refines decisions through a test-time retrieval and adaptation mechanism that consults the nearest k annotated neighbors from the training set to provide local evidence. Unlike previous retrieval methods, it operates directly on fused multimodal representations that are further geometry-conditioned, ensuring that the retrieved neighbors are more closely aligned with the final decision semantics rather than merely exhibiting surface-level similarity. Our contributions are summarized as:

- We propose a new paradigm for hateful video detection, formulating the task as evidence-grounded decision formation from both internal and external environments.
- We introduce the *P2C Generator*, which uses an MLLM to transform internal multimodal observations into structured cognitive evidence through perception-to-cognition reasoning.
- We propose the *Whitened Neighbor Interpolator*, a geometry-conditioned test-time retrieval module that refines ambiguous decisions with external annotated precedents that are more closely aligned with the final prediction space.

We conduct comprehensive experiments on three benchmarks across two languages, achieving state-of-the-art results with detailed analysis of both reasoning quality and retrieval effectiveness.

2 Method

Problem Formulation. Let q denote a video on online platforms. Each video q is represented by its multimodal content, including textual, visual, and audio modalities. For each video, we extract textual, visual, and audio information. For text, we directly use the video's metadata t, including title and video description. For visual content, we uniformly sample k frames from the video, denoted as $v = \{v_1, .., v_k\}$. For audio a, we obtain a global transcript t_a using automatic speech recognition (ASR). The task is to determine whether the video q contains hateful content by jointly modeling all modalities t, v, a.

Overview. The overall framework of our method is illustrated in Fig. 3. It follows the general process of human cognitive reasoning from both internal and external perspectives, progressing from perception to cognition to response, and is instantiated through three modules: (1) the *P2C Generator*, which employs the MLLM to internally examine the given video in depth, moving from global and local perception of the content to summarized cognitive evidence; (2) *Schema-guided Modality Routing*, a fusion module that models the interaction

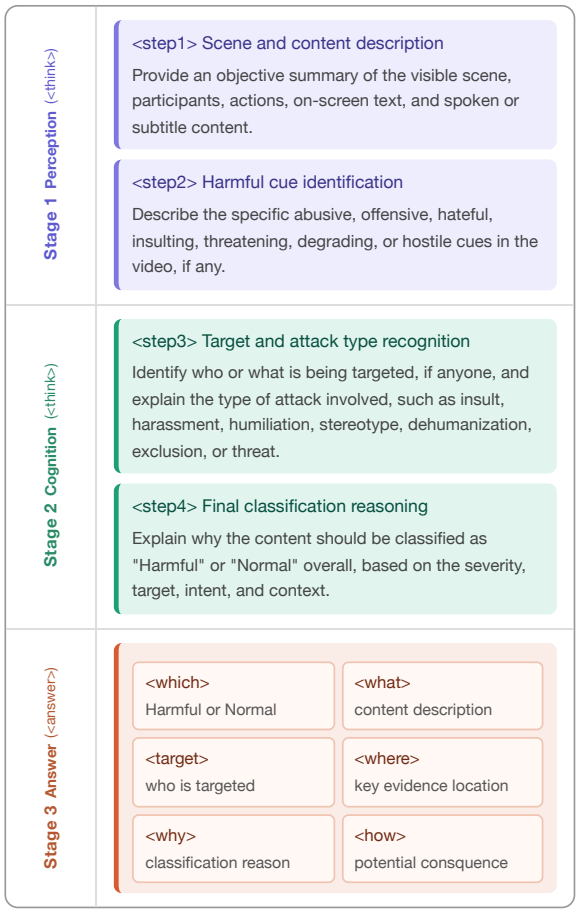

Figure 2: Prompt structure with key excerpts from the original prompt.

between multimodal video features and the cognitive schematic evidence produced by the MLLM in the latent space; and (3) the *Whitened Neighbor Interpolator*, which adopts a gradient-free test-time retrieval mechanism to externally identify precedent nearest neighbors with annotated ground truth and refine decisions on detected samples with reduced ambiguity.

2.1 P2C Generator

Most multimodal fusion-based methods directly map observable features to final judgments, skipping the crucial intermediate step of cognition. While some MLLM-based methods incorporate additional textual analysis, they still fail to fully unlock the reasoning capabilities of MLLMs, remaining largely at the surface level. Ideally, the modeling approach should mimic human annotators, who follow a complete process from perception to cognition to evidence-supported decision—consistent with cognitive appraisal theory. To this end, we propose a *P2C Generator* that incorporates a perception-to-cognition CoT to further exploit MLLM reasoning capabilities. As shown in Fig. 2, the MLLM is prompted to generate `<think>`...`</think>` and `<answer>`...`</answer>` tags, where it first progresses from perception to cognition and then summarizes all identified evidence.

Stage 1: Perception. Given a video t, t_a, v, a, humans first observe the video holistically. When signals of hateful content appear, they allocate more localized and focused attention to the relevant clips. Following this process, in Stage 1, which consists of Step 1 and Step 2, we expect the model to first form a global perception through *Scene and Context Description*, which provides an objective overview of the environment, participants, visible text, symbols, and overall activity in the video, and then conduct localized perception through *Hateful Cues Description*, which describes specific verbal, textual, visual, or symbolic content that may indicate hateful expression.

Stage 2: Cognition. After acquiring sufficient surface-level information, humans conduct reasoning based on the perceived cues and infer possible consequences. In hateful video assessment, this process moves from shallow to deep understanding: it first identifies who or what is being targeted, if anyone, and explains the type of attack involved, such as insult, harassment, or humiliation, which we term *Target and Attack Identification* (Step 3); it then performs *Contextual Harm and Norm Reasoning* (Step 4), which explains why the content should be considered hateful in context.

Stage 3: Answer. After completing the reasoning process, the MLLM is expected to provide a short structured summary related to the key evidential dimensions of hateful content in the video, including the classification (**which**), a description of the hateful content (**what**, w), the spatio-temporal-audio key evidence (**where**, e), the reason (**why**, r) for this classification, and **how** (o) it may promote hateful activity and cause harm.

Note that all three stages are completed within a single prompt call. In the subsequent modules, we use only the structurally defined fields marked by the mathematical symbols from the returned answer, as they provide a concise summary of the preceding thinking process while preserving all the key evidence we require.

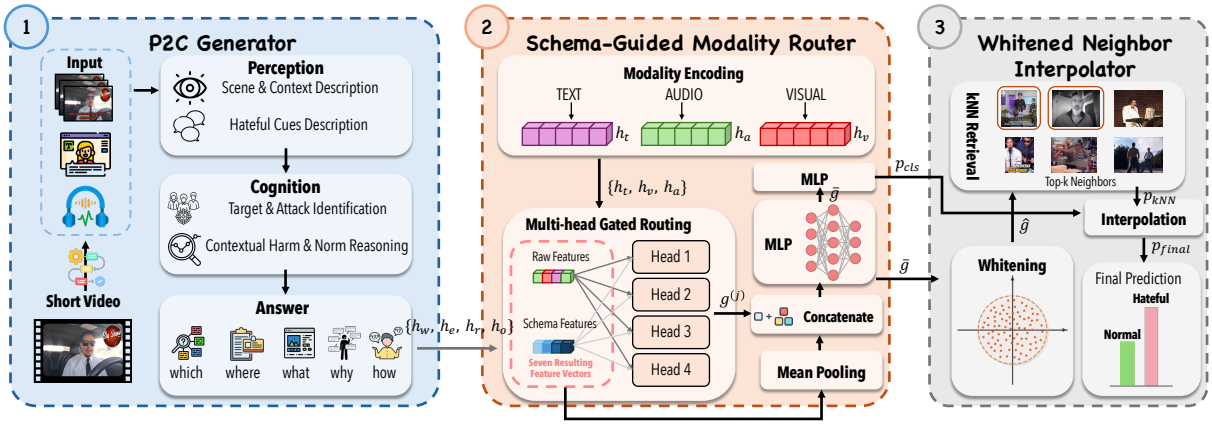

Figure 3: Overview of the HATELENS framework. Given a short video, (1) the CAT-guided dual-stage LLM pipeline first extracts an evidence ledger from multimodal content, then performs structured appraisal inference; (2) the appraisal-conditioned fusion module integrates raw media features with LLM-derived appraisal representations; (3) at inference time, the case-based retrieval module refines uncertain predictions by consulting similar labeled instances in a geometrically conditioned embedding space.

2.2 Schema-Guided Modality Router

Modality routing has been shown to be an effective fusion paradigm for multimodal feature integration. In this regard, we view the structural fields returned by the MLLM (i.e., w, e, r, and o) as a reasoning schema that can more effectively guide multimodal fusion.

We map each modality input into latent features using the corresponding off-the-shelf encoders, including a pre-trained language model (LM), visual transformer (VT), and audio transformer (AT), as:

$$\begin{aligned} \boldsymbol{h}_t &= \mathrm{LM}\left(t; t_a\right), \\ \boldsymbol{h}_v &= \mathrm{VT}(v), \\ \boldsymbol{h}_a &= \mathrm{AT}(a), \end{aligned} \tag{1}$$

where $\boldsymbol{h}_* \in \mathbb{R}^d$ denotes the extracted feature vector after applying mean pooling, and ; denotes concatenation. Accordingly, we also encode the textual structural fields separately using the same LM, resulting in $\boldsymbol{h}_{w/e/r/o}$.

Subsequently, we fuse the seven resulting feature vectors using a multi-head gated routing mechanism:

$$\begin{aligned} \alpha_m^{(j)} &= \frac{\exp(f_j(\mathbf{h}_m))}{\sum_{m' \in \mathcal{M}} \exp(f_j(\mathbf{h}_{m'}))}, \\ \boldsymbol{g}^{(j)} &= \sum_{m \in \mathcal{M}} \alpha_m^{(j)} \boldsymbol{h}_m, \end{aligned} \tag{2}$$

where $\mathcal{M} = \{t, v, a, w, e, r, o\}$. The final representation is formed by concatenating the outputs of all routing heads with a mean-pooled modality summary, and is then passed through a classification MLP:

$$\begin{aligned} \bar{\boldsymbol{g}} &= \mathrm{MLP}([\boldsymbol{g}^{(1)}; \ldots; \boldsymbol{g}^{(H)}]) \\ p_{cls} &= \mathrm{MLP}\left(\bar{\boldsymbol{g}}\right), \end{aligned} \tag{3}$$

where $\bar{\boldsymbol{g}}$ is the fused representation and $p_{cls} \in \mathbb{R}^2$ denotes the predicted probability distribution over the hateful and normal classes for the input video. The model is trained with the cross-entropy loss.

2.3 Whitened Neighbor Interpolator

Hate judgments are often inherently ambiguous: the same surface content may be hateful, satirical, educational, or counter-speech depending on context and framing. In the video setting, such expressions are even more diverse yet individually rare, resulting in a long-tail distribution. To address this, we propose leveraging training samples with similar fused features and annotated ground truth to sharpen the decision boundary. This design mirrors content moderation practice, where moderators learn general policies but consult prior precedent cases for difficult judgments (Klonick, 2018; Clarke et al., 2023). Specifically, we split this into two steps.

Geometric Conditioning. Previous methods (Lang et al., 2025) for multimodal retrieval typically operate directly on the initially extracted multimodal features with manually configured modality weights. This is problematic for two reasons. First, such designs are difficult to adapt to dynamically varying cases. Second, the initial features are not sufficiently discriminative for the final classification objective, which limits the usefulness of

the retrieved cases for informing decisions. This motivates us to instead leverage the fused penultimate representations $\bar{\boldsymbol{g}}$. However, these representations combine heterogeneous sources, including raw multimodal features and LLM-derived reasoning, which results in an anisotropic embedding space where distances become unreliable (Yi et al., 2025; Tsukagoshi and Sasano, 2025). To enable meaningful similarity-based retrieval, we apply covariance-based whitening:

$$\hat{\boldsymbol{g}} = \frac{(\bar{\boldsymbol{g}} - \boldsymbol{\mu})\mathbf{W}_{\text{whiten}}}{\|(\bar{\boldsymbol{g}} - \boldsymbol{\mu})\mathbf{W}_{\text{whiten}}\|_2}, \tag{4}$$

where $\boldsymbol{\mu}$ is the training set mean and $\mathbf{W}_{\text{whiten}}$ is derived from shrinkage-regularized covariance estimation (Yi et al., 2025). This conditioning ensures that similarity computations reflect genuine semantic proximity rather than artifacts of the heterogeneous feature geometry.

Retrieval and Interpolation. At inference, we store whitened penultimate representations $\{\hat{\boldsymbol{h}}_i\}_{i=1}^{N}$ from the training set along with their labels. For a test sample with representation $\hat{\boldsymbol{g}}_{\text{test}}$, we retrieve its k nearest neighbors and compute a soft label distribution:

$$p_{\text{kNN}}(y = c) = \frac{\sum_{i \in \mathcal{N}_k} \mathbb{1}[y_i = c] \cdot \kappa(\hat{\boldsymbol{g}}_{\text{test}}, \hat{\boldsymbol{g}}_i)}{\sum_{i \in \mathcal{N}_k} \kappa(\hat{\boldsymbol{g}}_{\text{test}}, \hat{\boldsymbol{g}}_i)}, \tag{5}$$

where $\mathcal{N}_k$ denotes the k nearest neighbors and $\kappa(\cdot, \cdot) = \exp(\text{sim}(\cdot, \cdot))$ is a similarity kernel, which enhances weights from more similar neighbors. The final prediction interpolates the classifier logits with the retrieval distribution:

$$p_{\text{final}} = (1 - \alpha) \cdot p_{cls} + \alpha \cdot p_{\text{kNN}}, \tag{6}$$

where α is the trade-off parameter.

3 Experiments

3.1 Experimental Setup

Datasets. We evaluate our method on three hateful video detection benchmarks: HateMM (Das et al., 2023), MultiHateClip-Youtube (MHClip-Y), and MultiHateClip-Bilibili (MHClip-B). We adopt the official data splits from Lang et al. (2025), and Table 1 summarizes the dataset statistics. Following prior work, MHClip-Y and MHClip-B are evaluated under a binary setting (Normal vs. Offensive+Hateful). Results for three-way classification are provided in the Appendix.

Table 1: Dataset statistics.

		Split			Class Dist.		
Dataset	**Total**	**Train**	**Val**	**Test**	**Norm.**	**Off.**	**Hat.**
HateMM	1,066	744	107	215	639	—	427
MHClip-Y	891	618	91	182	601	218	72
MHClip-B	897	633	88	176	605	180	112
ImpliHateVid	2,009	1,283	325	401	1,000	—	1,009

Baselines. We compare against baselines from three categories: (1) **Unimodal**: BERT (Devlin et al., 2019), ViViT (Dosovitskiy et al., 2021), MFCC (Davis and Mermelstein, 1980); (2) **Multimodal fusion**: Pro-Cap (Cao et al., 2023), HateMM (Das et al., 2023), MHCL (Wang et al., 2024a), MoRE (Lang et al., 2025), ImpliHateVid (Rehman et al., 2025), CMFusion (Zhang et al., 2024); (3) **MLLM-based**: MiniCPM-V, LLaVA-OV (Li et al., 2024), Qwen2-VL (Wang et al., 2024b), HVGuard (Jing et al., 2025). Details of baselines are in Appendix.

Metrics. Following previous work (Lang et al., 2025; Jing et al., 2025), we report Accuracy (**ACC**), Macro-F1 (**M-F1**), Macro-Precision (**M-P**), and Macro-Recall (**M-R**). The best checkpoint selected on the validation set is then evaluated on the test set for performance reporting. All results are averaged over 20 random seeds.

Implementation Details. We use `GPT-5.4-nano` as the MLLM backbone. Text features are encoded with `BERT` (Devlin et al., 2019), visual features with `ViT` (Dosovitskiy et al., 2021), and audio features with `WavLM` (Chen et al., 2022); all encoders are kept frozen. The fusion module uses $H = 4$ routing heads with hidden dimension $h = 192$. For retrieval, we set $k = 15$ and $\alpha = 0.2$. Further implementation details are provided in Appendix **??**.

3.2 Comparison with SOTA

Table 2 presents the main results, from which we highlight three key observations. **(O1): Multimodal fusion outperforms unimodal approaches**, as hateful expression may emerge across different modalities that single-modality models cannot capture. **(O2): Incorporating MLLMs into multimodal modeling further improves performance.** MLLM-based methods (e.g., HVGuard) consistently outperform feature-fusion baselines, and our method extends this advantage further by guid-

Table 2: Comparison with state-of-the-art methods on four hateful video detection benchmarks. Best results in **bold**, second best underlined. ↑ indicates the improvement of HATELENS over the best baseline. a_{-b} denotes $a \times 10^{-b}$;

	HateMM				MHClip-Y				MHClip-B				ImpliHateVid			
Method	ACC	M-F1	M-P	M-R	ACC	M-F1	M-P	M-R	ACC	M-F1	M-P	M-R	ACC	M-F1	M-P	M-R
Unimodal Methods																
BERT	69.1	63.7	70.1	64.0	65.5	49.1	55.2	52.2	72.5	67.7	68.4	62.8	75.2	75.2	75.3	75.2
ViViT	68.2	66.7	66.8	66.6	61.4	61.4	62.2	61.1	71.0	66.1	66.6	65.8	76.5	76.5	76.7	76.5
MFCC	65.4	60.3	64.1	60.7	46.5	47.2	58.8	52.2	63.1	52.5	54.1	53.0	70.0	70.0	70.1	70.0
Multimodal Fusion Methods																
Pro-Cap	64.5	63.3	63.4	63.2	70.1	66.3	66.3	66.3	72.5	66.8	66.1	68.3	82.3	82.3	82.4	82.3
HateMM	76.0	72.8	77.9	72.0	71.5	63.2	68.3	62.6	71.0	61.8	66.5	61.4	76.6	76.5	76.8	76.6
MHCL	77.4	76.5	77.5	76.6	65.4	65.4	67.2	64.9	76.5	73.1	73.2	73.0	85.2	85.2	85.4	85.2
MoRE	83.4	82.4	81.8	83.3	77.5	75.2	75.7	74.8	78.5	74.8	75.7	74.1	84.8	84.7	85.4	84.8
ImpliHateVid	82.8	82.0	82.1	82.0	76.1	71.7	71.6	71.8	78.3	72.8	74.4	71.8	87.5	87.5	87.6	87.5
CMFusion	75.6	74.1	75.3	73.7	72.2	68.8	69.5	68.2	73.8	70.1	71.0	69.4	68.5	68.4	68.9	68.6
Large Vision-Language Models																
MiniCPM-V	72.4	72.3	77.8	76.4	69.1	67.4	69.3	67.4	73.5	73.2	72.0	73.0	–	–	–	–
LLaVA-OV	75.6	75.6	77.9	78.3	73.7	70.5	70.5	66.8	75.2	70.2	71.4	70.3	–	–	–	–
Qwen2-VL	73.7	73.7	78.1	77.3	70.5	66.8	66.8	66.7	73.3	74.0	73.9	72.9	–	–	–	–
HVGuard	87.9	87.5	87.3	87.6	79.8	76.2	77.0	75.6	79.6	75.6	76.0	75.3	86.0	86.0	86.1	86.0
HATELENS	**91.6**	**91.4**	**91.0**	**91.9**	**87.1**	**84.0**	**85.9**	**82.6**	**89.2**	**85.6**	**92.0**	**82.5**	**94.3**	**94.3**	**94.4**	**94.3**
↑ *Improv.*	*+3.7*	*+3.9*	*+3.7*	*+4.3*	*+7.3*	*+7.8*	*+8.9*	*+7.0*	*+9.6*	*+10.0*	*+16.0*	*+7.2*	*+6.8*	*+6.8*	*+6.8*	*+6.8*
p-val	3.2_{-6}	1.8_{-6}	7.4_{-5}	9.1_{-7}	5.6_{-6}	8.3_{-8}	2.1_{-7}	4.7_{-7}	1.4_{-6}	3.7_{-8}	6.2_{-9}	2.8_{-7}	8.9_{-6}	6.3_{-6}	2.1_{-5}	7.5_{-6}

ing the MLLM through a structured perception-to-cognition process that explicitly models evidence, targets, and contextual harm. **(O3): HATELENS achieves the strongest performance across all three datasets on all metrics.** Compared with HVGuard, HATELENS yields absolute accuracy gains of 3.7%, 7.3%, 9.6%, and 6.8% respectively, while its higher precision and recall indicate more hateful content detected with fewer false positives. These results validate our paradigm of first examining videos through progressively deeper perception and cognition, then consulting external cases to support the final decision.

3.3 Ablation Study

Table 3: Ablation study (%). $\overline{\Delta}$: mean ACC drop across four datasets.

	HateMM		MH-Y		MH-B		ImpliHV		
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	$\overline{\Delta}$
Full	**91.6**	**91.4**	**87.1**	**84.0**	**89.2**	**85.6**	**94.3**	**94.3**	—
P2C Generator									
−*what*	87.4	86.7	82.1	77.0	86.6	83.0	91.3	91.3	−3.7
−*where*	87.9	87.3	82.8	79.8	84.7	83.1	92.8	92.8	−3.5
−*why*	87.6	87.0	82.2	78.4	85.4	84.0	92.3	92.3	−3.7
−*how*	87.1	86.2	81.9	76.6	84.7	82.2	92.8	92.8	−3.9
Percep. only	87.9	87.4	80.9	75.1	83.5	80.9	91.8	91.8	−4.5
Cogn. only	86.0	85.5	80.2	74.3	82.1	84.1	89.5	89.5	−6.1
HVGuard CoT	85.1	84.3	79.1	73.2	78.3	73.2	89.3	89.3	−7.6
Schema-Guided Router									
MoE fusion	85.6	84.8	77.9	70.7	81.5	79.1	91.3	91.3	−6.5
HVGuard fus.	87.0	86.2	80.4	75.1	80.9	76.9	92.0	92.0	−5.5
Whitened Neighbor Interp.									
−Interp.	87.0	86.4	80.4	75.4	86.0	82.9	91.0	91.0	−4.5
−whitening	88.4	87.7	81.0	74.9	85.4	82.2	93.5	93.5	−3.5
Pre-fus. retr.	85.6	84.4	82.8	77.2	82.2	73.7	91.0	91.0	−5.2

Table 3 reports the ablation results across three benchmarks. **Effect of P2C Generator.** Removing individual answer fields (-*what*, -*where*, -*why*, -*how*) consistently degrades performance, with *what* and *how* showing the largest impact, indicating that both content description and harm contextualization are essential. Using only the perception stage (Percep. only) or only the cognition stage (Cogn. only) underperforms the full answer, confirming their complementary roles. Replacing our P2C prompt with the HVGuard CoT prompt (Jing et al., 2025) while keeping the same fusion and retrieval modules causes a substantial drop, demonstrating that structured perception-to-cognition decomposition produces more informative representations than a describe-analyze-judge pipeline. **Effect of Schema-Guided Router.** Substituting our fusion module with those from MoRE and HVGuard leads to notable degradation, which we attribute partly to the optimization difficulty and expert collapse risks of MoE-based architectures, whereas our router is more stable and effectively integrates multimodal features with LLM-derived reasoning. **Effect of Whitened Neighbor Interpolator.** Removing the interpolator (-Interp.) causes a clear drop, confirming the value of consulting external cases. Geometric conditioning (-whitening) is also important, as the fused representations are anisotropic and less

Table 4: Explanation quality evaluated by GPT-4o on 100 MHClip-Y samples (5-point Likert). [Mock data.]

Method	Info.	Sou.	Per.	Rea.	Flu.	Avg.
HVGuard	3.42	3.51	3.28	4.15	4.20	3.69
RAMF	3.38	3.45	3.35	4.02	4.21	3.63
MARS	3.41	3.34	3.57	3.96	4.18	3.69
Ours	**4.12**	**4.05**	**3.92**	**4.28**	**4.21**	**4.12**

reliable without this step. Retrieval based on initial multimodal features (Pre-fus. retr.) also underperforms, indicating that such a mechanism struggles with the complex and dynamic nature of this task.

3.4 P2C CoT Analysis

An effective MLLM-based CoT should exhibit two properties: (1) the output reasoning should provide informative evidence that aids human decision-making, and (2) the CoT design should generalize across different MLLMs with consistent performance gains. We evaluate both through quantitative analysis, comparing against HVGuard (Jing et al., 2025), MARS (Yang et al., 2026) and RAMF (Yang et al., 2025).

Explanation Quality. We randomly sample 100 videos of each dataset and evaluate the reasoning outputs from three methods using GPT-4o as an expert judge across five criteria: (1) *Informativeness*: whether the explanation provides background and context beyond surface description; (2) *Soundness*: whether the reasoning is valid and logically coherent; (3) *Persuasiveness*: whether the conclusion is convincing; (4) *Readability*: whether the text follows proper grammar and structure; and (5) *Fluency*: whether the explanation flows smoothly with well-connected ideas. Each criterion is rated on a 5-point Likert scale. For detailed prompt, refer to Appendix.

As shown in Table 4, our P2C reasoning achieves the highest scores across all five criteria. This suggests that the structured perception-to-cognition decomposition produces more thorough and well-grounded explanations compared to the describe-analyze-judge paradigm.

Generalizability across MLLMs. To assess whether our CoT design transfers across different MLLMs, we apply the P2C prompt to three models of varying scale: GPT-5.4-nano, Qwen2-VL (Wang et al., 2024b), and LLaVA-OV (Li et al., 2024). As shown in Figure 4, applying our P2C CoT consistently improves each MLLM's performance . This

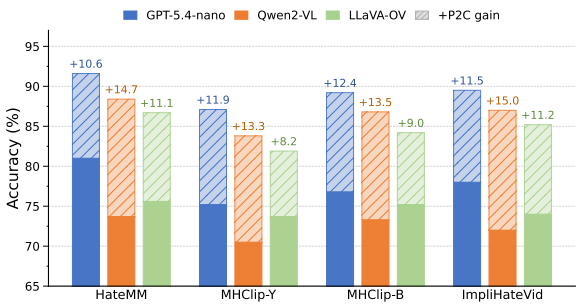

Figure 4: Generalizability of P2C CoT across different MLLMs. Applying our CoT design consistently improves performance over each MLLM's default prompting across all three datasets. [Mock data — to be replaced with real experiments.]

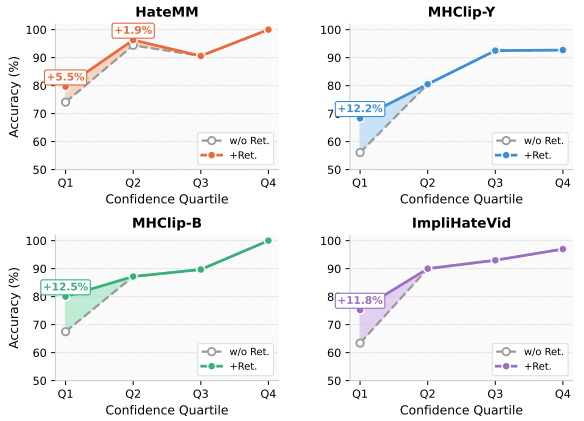

Figure 5: Interpolator gain by classifier confidence quartile across three datasets. Gains are concentrated in low-confidence bins.

confirms that the perception-to-cognition decomposition is a generalizable design principle.

3.5 Interpolator Analysis

Correcting Hard Samples. To validate that the Whitened Neighbor Interpolator addresses the decision challenge on ambiguous content, we analyze *where* interpolator helps by binning test samples into quartiles based on the classifier's prediction confidence. As shown in Figure 5, interpolator gains are overwhelmingly concentrated in low-confidence bins. This pattern is consistent across all three datasets, confirming that retrieval acts as targeted decision support for ambiguous cases rather than a uniform post-hoc adjustment.

Enhancement of Transferability. Hateful video detection models are often trained on one platform but deployed on another. To evaluate cross-domain robustness, we conduct exhaustive transfer experiments: for each of the three datasets, we train

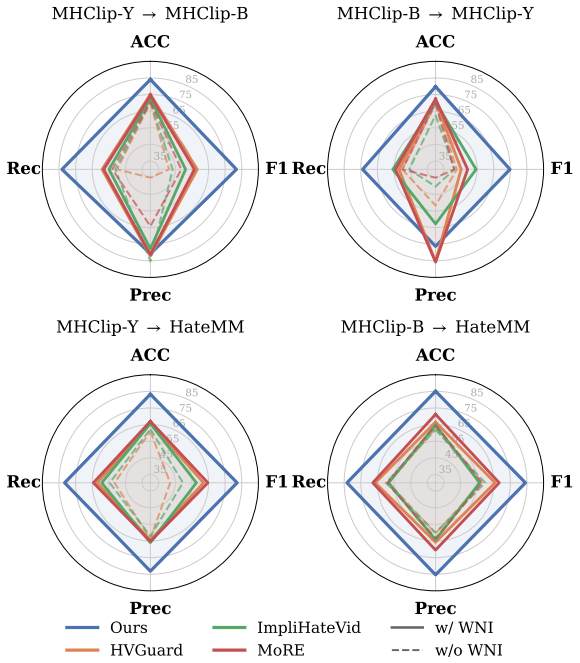

Figure 6: Cross-dataset transferability comparison. Solid/dashed lines show baselines with and without our Whitened Neighbor Interpolator.

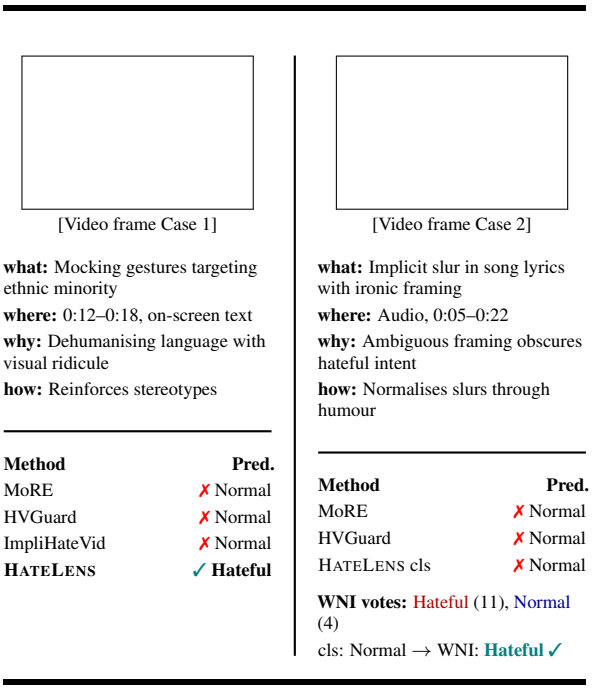

Figure 7: Case study layout. Two cases are separated by a solid vertical rule for maximum compilation stability and a clean academic look.

on one (source) and directly evaluate on the test set of another (target). We mainly focus on the performance of the baselines after plug-and-play integration of our interpolator module, where training instances from the source domain are used as the neighbor bank.

Figure 6 shows the transfer results for all six source→target directions. We observe that after applying our interpolator module, the performance of all baselines improves substantially. This provides strong evidence that the interpolator can serve as an effective test-time adaptation module across varied contexts. Furthermore, when combined with P2C reasoning, our method achieves the strongest performance, benefiting from the complementary strengths of both components and yielding a more transferable decision boundary.

3.6 Case Study

Figure **??** presents the predictions for two representative cases from the baselines and our method. In the first case, the hateful expression is relatively implicit. As a result, the other three baselines make incorrect predictions and fail to identify it as hateful. Methods without MLLMs do not provide explanations, which further increases decision difficulty, while MLLM-based methods remain at the level of surface multimodal analysis and produce judgments without collecting intermediate cognitive evidence. In contrast, our method fully exploits the reasoning capability of the MLLM, progressing from perception to cognitive evidence and finally to judgment, thereby successfully detecting this case. The second case is even more challenging, and all methods, including ours, initially classify it as non-hate. However, because our method is equipped with the interpolator with external retrieval at inference time, it can effectively leverage similar samples with annotated ground truth and ultimately correct the prediction.

4 Related Work

Our work is closely related to video-based hate speech detection. Early hate speech detection research mainly focused on text-based settings, typically using traditional machine learning classifiers (Davidson et al., 2017). With the rise of deep learning, increasing attention has been devoted to hate detection in multimodal settings, from the introduction of image-text meme benchmarks (Kiela et al., 2020) to more recent studies on hateful video detection (Das et al., 2023; Wang et al., 2024a; Rehman et al., 2025). For videos, most existing work extends conventional visual-textual feature fusion to three modalities, i.e., visual, textual, and audio fusion, through different fusion mechanisms

(Céspedes-Sarrias et al., 2025; Zhang et al., 2024; Lang et al., 2025; Yue et al., 2025; Rehman et al., 2025). With the emergence of MLLMs, some studies have begun to explore different prompting techniques, either by generating rationales to assist the training of downstream conventional models (Jing et al., 2025; Cao et al., 2023), or by directly performing zero-shot inference with MLLMs (Yang et al., 2025). However, most existing methods share two common limitations: they do not provide a well-grounded understanding of video content, but mainly model *what* is in the video, and they lack an effective mechanism for handling ambiguous cases involving implicit and diverse hateful expression. In this work, we first propose to internally understand videos through a perception-to-cognition reasoning process, and then resolve ambiguity by consulting annotated precedents that provide meaningful local structural evidence.

5 Conclusion

We presented HATELENS, a framework that reframes hateful video detection as an appraisal problem. By grounding LLM-based analysis in Cognitive Appraisal Theory, our dual-stage pipeline produces structured intermediate representations—relational meaning and alternative appraisals—that capture the interpretive judgments necessary for distinguishing hateful endorsement from reporting, satire, and counterspeech. A case-based retrieval module further refines uncertain predictions by consulting similar labeled instances, providing targeted decision support for ambiguous content.

Beyond overall accuracy improvements, our most consequential practical result is the substantial reduction in false positives on benign content containing hate-adjacent language: HATELENS reduces FPR by 50–60% on quoted hate and counterspeech compared to existing methods. This has direct implications for content moderation, where false positives on legitimate speech are costly. We believe the appraisal-centered perspective offers a productive direction for building more nuanced and trustworthy hate detection systems.

Limitations

Our framework relies on a commercial LLM (GPT-4.1-nano) for the appraisal stage, which introduces API costs and latency. The two-call pipeline approximately doubles inference time compared to single-call approaches. Additionally, our case-based retrieval module requires storing training set embeddings, though the storage overhead is minimal (<1MB for all datasets). The framework has been evaluated on English and Chinese datasets; its effectiveness on other languages and cultural contexts remains to be verified.

Ethics Statement

This work involves detecting hateful content in online videos, which necessarily requires engaging with offensive material. All datasets used are publicly available and were collected with appropriate ethical oversight by their original creators. We note several ethical considerations: (1) Hate speech annotations are inherently subjective and may reflect annotator biases; our method inherits these biases through training data. (2) Automated detection systems risk both false positives (censoring legitimate speech) and false negatives (failing to flag harmful content), with disparate impacts across communities. (3) Our system should be used as a *decision-support tool* alongside human review, not as an autonomous moderator. (4) The LLM component may generate harmful text during analysis; we process outputs programmatically without displaying them to end users. (5) Cross-cultural hate norms vary; performance on Chinese content may not reflect all cultural contexts within Chinese-speaking communities.

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A Baseline Descriptions

We compare HATELENS against three categories of baselines.

Unimodal Baselines.

- **BERT** (**?**) is a bidirectional transformer pre-trained via masked language modelling. We fine-tune `bert-base-uncased` on the concatenation of the video title and ASR transcript.
- **ViViT** (**?**) is a pure-transformer model for video classification that extracts spatio-temporal tokens from uniformly sampled frames. We use the factorised encoder variant pre-trained on Kinetics-400 and fine-tune it on the visual stream of each dataset.
- **MFCC** (**?**) is a classical audio feature based on Mel-frequency cepstral coefficients. We extract 40-dimensional MFCC features, compute mean and standard deviation pooling, and train an MLP classifier on the resulting representation.

Multimodal Fusion Baselines.

- **Pro-Cap** (Cao et al., 2023) prompts a frozen vision-language model with hateful-content-related questions in a zero-shot VQA manner and uses the answers as informative captions for a downstream text classifier. We adapt this pipeline to video by applying it to a representative frame.
- **HateMM** (Das et al., 2023) encodes text with BERT, visual content with a 3D-CNN, and audio with a pre-trained audio model, then fuses the three streams via cross-modal attention.
- **MHCL** (Wang et al., 2024a) is the baseline introduced alongside the MultiHateClip benchmark. It aligns all three modality representations through contrastive learning objectives for cross-lingual hate-video detection.
- **MoRE** (Lang et al., 2025) is a retrieval-augmented mixture-of-experts model that employs modality-specific experts and retrieves annotated neighbours using early-fusion representations with manually specified modality weights.
- **ImpliHateVid** (Rehman et al., 2025) proposes a two-stage contrastive learning framework for implicit hate video detection. The first stage aligns modality representations, and the second stage contrasts hateful with non-hateful examples to improve detection of subtle cues.
- **CMFusion** (**?**) applies separate attention mechanisms along both the channel and modality dimensions for multimodal feature fusion in hate video detection.

MLLM-based Baselines.

- **MiniCPM-V** (**?**) is a series of efficient MLLMs designed for end-side deployment. We use the `MiniCPM-Llama3-V-2.5` variant and apply it in a zero-shot prompting setting with a representative frame grid and video metadata.
- **LLaVA-OV** (**?**) (LLaVA-OneVision) supports single-image, multi-image, and video understanding via a unified architecture. We prompt it in a zero-shot chain-of-thought setting to describe the content and produce a binary judgment.
- **Qwen2-VL** (**?**) encodes images and videos at their native resolution via a Naive Dynamic Resolution module and Multimodal Rotary Position Embedding (M-RoPE). We evaluate it under the same zero-shot prompting setting as LLaVA-OV.
- **HVGuard** (**?**) prompts an MLLM with a describe-then-analyze chain-of-thought to generate rationales, and then fine-tunes a downstream fusion classifier on the rationales together with multimodal features.

All baselines strictly follow the released data splits from Lang et al. (2025) and Rehman et al. (2025) and adopt their optimal settings.

B Additional Experiments

B.1 Three-Way Classification

Table 5 reports results under the full three-way classification setting (Normal / Offensive / Hateful) on MHClip-Y and MHClip-B. This is a strictly harder task than the binary setting in the main paper, as the model must additionally distinguish offensive content from hateful content, a boundary that is inherently subjective and often ambiguous. HATELENS achieves the best performance on all metrics across both datasets. The gains in M-P are particularly large, suggesting that both the structured P2C reasoning and the Whitened Neighbor Interpolator contribute to suppressing false positives: the former grounds predictions in explicit cognitive evidence about targets, intent, and contextual harm, while the latter retrieves annotated precedents that help resolve ambiguous cases near the decision boundary. The improvement in M-F1 over the best baseline is more modest, reflecting the inherent difficulty of the offensive class, which is small and semantically close to both Normal and Hateful.

B.2 Hyperparameter Sensitivity

Figure 8 reports the accuracy of HATELENS as a function of the two key hyperparameters of the Whitened Neighbor Interpolator: the number of retrieved neighbours $k \in \{1, 3, 5, 10, 15, 25, 50\}$ and the interpolation weight $\alpha \in \{0.1, 0.15, 0.2, 0.3, 0.4, 0.5\}$. Across all four datasets the method is broadly stable. Performance improves as more neighbours are included, and is further boosted when α is set to a moderate value that balances the parametric

Table 5: Three-way classification results (Normal / Offensive / Hateful) on MHClip-Y and MHClip-B. Best results in **bold**, second best underlined. ↑ indicates the improvement of HATELENS over the best baseline.

	MHClip-Y (3-way)				MHClip-B (3-way)			
Method	ACC	M-F1	M-P	M-R	ACC	M-F1	M-P	M-R
Unimodal Methods								
BERT	69.8	41.9	38.9	40.1	69.3	34.5	36.9	36.5
ViViT	73.0	40.5	57.9	38.3	70.1	27.5	49.5	34.2
MFCC	69.9	27.4	24.5	33.1	70.1	27.5	23.7	33.2
Multimodal Fusion Methods								
Pro-Cap	73.6	46.9	58.4	44.1	68.8	43.5	48.9	41.7
HateMM†	73.0	47.0	59.1	46.6	70.1	44.7	48.6	43.4
MHCL	73.6	53.9	51.8	47.8	72.6	51.6	49.7	46.5
MoRE	71.8	39.7	41.6	40.6	72.0	33.7	44.1	36.7
CMFusion†	73.6	47.9	59.9	44.8	71.3	47.1	50.5	43.0
ImpliHateVid	69.9	27.4	23.3	33.3	73.3	45.9	42.8	50.2
Large Vision-Language Models								
HVGuard	73.6	41.8	66.3	41.2	71.3	36.7	47.3	38.0
HATELENS	**79.1**	**57.0**	**86.2**	**52.4**	**77.1**	**58.2**	**63.5**	**55.5**
↑ *Improv.*	*+5.5*	*+3.1*	*+19.9*	*+4.6*	*+3.8*	*+6.6*	*+16.2*	*+5.3*

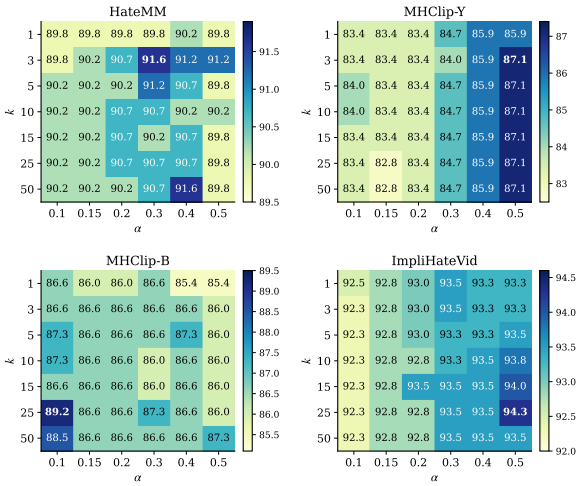
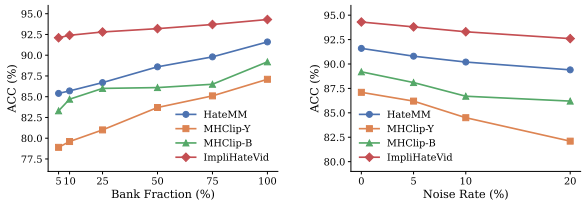


Figure 8: Sensitivity of test accuracy to the number of retrieved neighbours k and the interpolation weight α across all four datasets. Darker cells indicate higher accuracy.



(a) Bank size. (b) Label noise.

Figure 9: Retrieval robustness analysis.

classifier signal with the local retrieval evidence. We adopt k=15 and α=0.2 as defaults.

B.3 Retrieval Robustness

Figure 9a varies the fraction of the training set used to populate the retrieval bank from 5% to 100%. Performance improves monotonically and plateaus around 75%–100%, confirming that the interpolator benefits from a larger and more diverse evidence pool. Notably, even with only 25% of the bank the model retains most of the improvement from full-bank retrieval, showing robustness to partial-bank settings.

Figure 9b evaluates robustness by randomly flipping a fraction of training-set labels (0%–20% noise rate) before populating the retrieval bank. Accuracy degrades gracefully: even at 20% label noise the model retains strong performance across all datasets. This is because the whitening step geometrically conditions the embedding space, which reduces the influence of outlier neighbours that are likely to correspond to mislabelled instances.

B.4 Router Weight Analysis

Figure 10 visualizes the learned routing weights across the H=4 attention heads for each of the seven modality inputs (Txt, Aud, Frm, What, Tgt, Whr, Why, How) on all four datasets. The weights are fairly distributed across modalities, indicating that all inputs contribute to the fused representation. On ImpliHateVid, frame features (Frm) and content-description schema fields (What) receive notably higher average weight, consistent with the

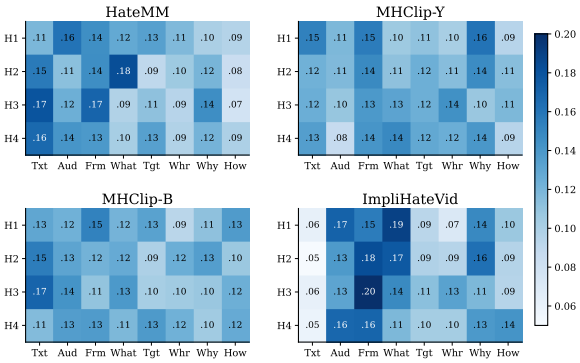


Figure 10: Mean routing weights per head and modality input on the four test sets.

visually implicit nature of that dataset. Across datasets, the schema-derived fields (What, Whr, Why, How) collectively account for roughly 35–45% of the total routing mass, confirming that the P2C reasoning schema provides complementary discriminative signal beyond the raw multimodal features.

B.5 Full Cross-Dataset Transfer Results

Figure 11 extends the cross-dataset transfer experiment in Section 3.5 of the main paper to all twelve source→target directions, including transfers involving the ImpliHateVid dataset. The Whitened Neighbor Interpolator (w/ WNI) consistently improves over the no-retrieval baseline (w/o WNI) across all methods and directions. HATELENS achieves the best or second-best result in all twelve settings, with the largest gains on directions involving ImpliHateVid as the target domain, where the implicit nature of the hate expressions makes retrieval-based disambiguation most valuable.

B.6 Efficiency Analysis

Table 7 summarises the computational profile of HATELENS. The fusion module contains only 1.52M trainable parameters, fewer than both HVGuard (2.52M) and ImpliHateVid (6.37M). Training converges in under 10 seconds, inference latency stays below 1ms per sample, and the retrieval step adds at most 0.12ms overhead. The feature bank requires no more than 321KB of storage across all datasets, confirming that the Whitened Neighbor Interpolator introduces negligible cost in practice.

B.7 Feature Visualization

Figure 12 visualizes the fused representations after the Schema-guided Modality Router via t-SNE.

Table 6: GPT-4o judge prompt for explanation quality evaluation.

System: You are an expert annotator evaluating the quality of machine-generated explanations for hateful video detection.

User: Given the following video metadata, frames and transcript, and the reasoning produced by a detection system, please rate the explanation on five criteria, each on a scale of 1 (very poor) to 5 (excellent):
1. Informativeness: Does the explanation provide background, context, or cultural knowledge beyond surface-level description?
2. Soundness: Is the reasoning logically coherent and free of contradictions?
3. Persuasiveness: Is the conclusion convincing given the provided evidence?
4. Readability: Does the text follow proper grammar and sentence structure?
5. Fluency: Does the explanation flow smoothly with well-connected ideas?
Return your ratings as a JSON object: `{"informativeness": X, "soundness": X, "persuasiveness": X, "readability": X, "fluency": X}`

Across all four datasets, HATELENS produces markedly better-separated clusters than the baselines, with Normal and Hateful samples forming more compact and distinct groups. This separation stems from the P2C Generator, which supplies structured cognitive evidence about targets, intent, and contextual harm; the Schema-guided Modality Router then models the interaction between this evidence and the raw multimodal features in the latent space, yielding representations that reflect the semantic structure of hateful content rather than surface-level feature correlations. Notably, the advantage is most pronounced on MHClip-B and ImpliHateVid, where implicit and diverse expressions render competing methods' clusters heavily entangled. The already-discriminative fused representations shown here are further geometry-conditioned by the whitening step in the Whitened Neighbor Interpolator, which corrects the anisotropy introduced by heterogeneous feature sources and ensures that similarity-based retrieval operates in a metrically meaningful space, enabling the kNN search to identify semantically relevant precedents rather than spurious nearest neighbours, and ultimately contributing to the significant performance gains reported in Table 2.

B.8 Modality Robustness

Figures 13–15 evaluate robustness by randomly corrupting one modality (Text, Audio, or Visual)

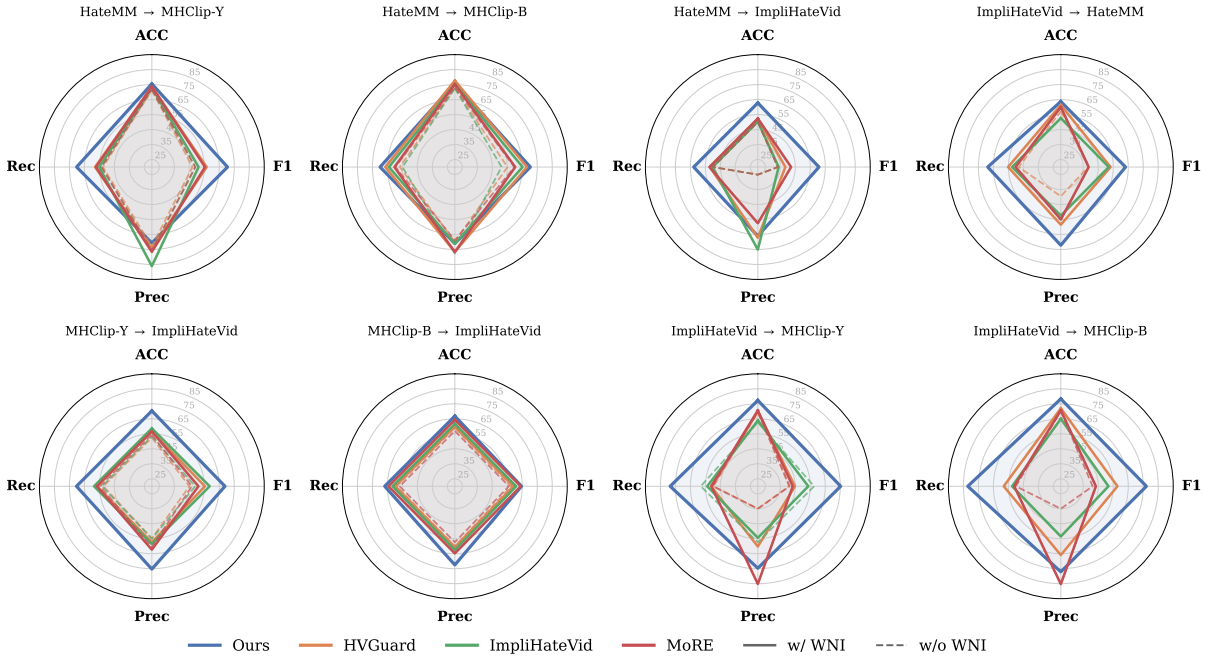


Figure 11: Full cross-dataset transfer results across all twelve source→target pairs. Radar plots show ACC, M-F1, M-P, and M-R. For each direction the solid coloured lines correspond to HATELENS (blue), HVGuard (orange), ImpliHateVid (green), and MoRE (red); the dark/dashed grey lines show the same baselines augmented with our Whitened Neighbor Interpolator (w/ WNI and w/o WNI respectively).

Table 7: Efficiency comparison. All experiments run on a single NVIDIA RTX 5090 GPU.

		Training Time (s)				
Method	**Params**	**HMM**	**MH-Y**	**MH-B**	**IHV**	**Infer. (ms)**
MoRE	0.42M	8.4	6.9	8.0	15.3	0.14
HVGuard	2.52M	10.9	8.2	8.0	18.1	0.38
ImpliHateVid	6.37M	10.0	8.5	9.4	19.2	0.43
HATELENS	**1.52M**	9.2	7.5	7.4	17.3	0.51

at rates of 25%–100% under three settings: train-time only, test-time only, and both simultaneously. Specifically, text corruption replaces sample tokens with random words, audio corruption substitutes the waveform of sampled clips with Gaussian noise, and visual corruption randomly masks or shuffles sampled frames. HATELENS consistently maintains the highest accuracy across all corruption settings, modalities, and datasets, with a notably flatter degradation curve compared to all baselines. This robustness stems from two complementary components. The P2C Generator produces structured cognitive evidence grounded in explicit reasoning about targets, intent, and contextual harm, which means HATELENS does not over-rely on any single raw modality stream; even when one modality is heavily corrupted, the P2C schema provides a stable signal that the Schema-guided Modality Router can leverage to compensate, a property that pure feature-fusion baselines fundamentally lack. The margin over MLLM-based baselines is particularly large under text corruption, where competing methods degrade more sharply because they rely on surface-level textual analysis rather than higher-level semantic judgments. Beyond the parametric model, the Whitened Neighbor Interpolator provides an additional layer of robustness at test time: by consulting geometrically well-conditioned nearest neighbours from the training bank, it can correct uncertain predictions even when the input representation is partially distorted by corruption, further stabilizing performance at high corruption rates.

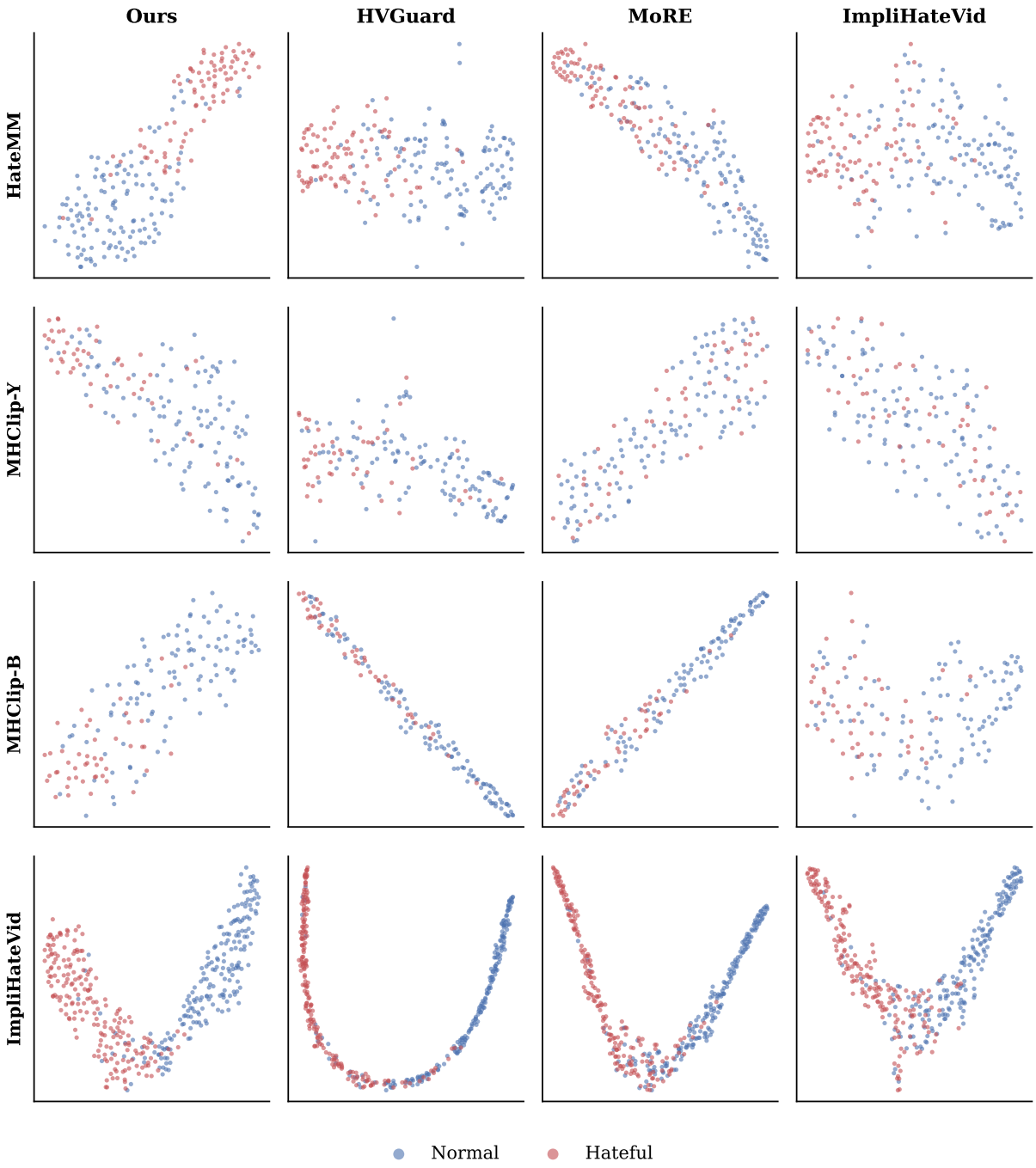


Figure 12: t-SNE visualization of learned representations on the test sets of all four datasets. Rows correspond to datasets; columns correspond to methods. Blue and red dots denote Normal and Hateful samples respectively.

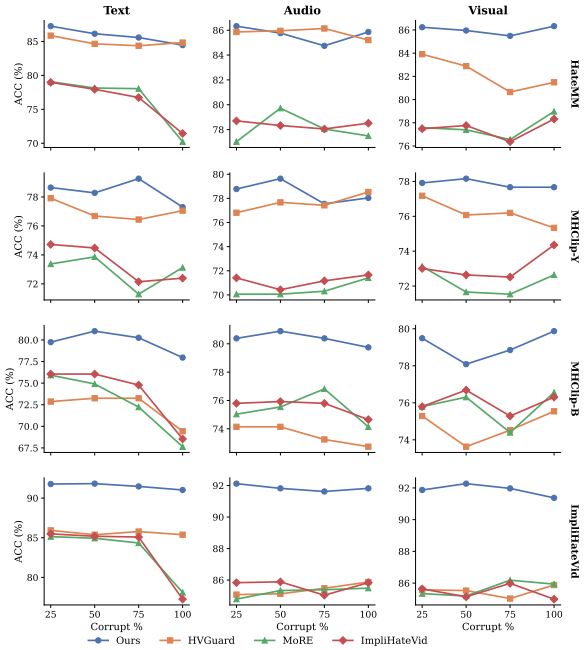


Figure 13: Modality robustness under train-time corruption. Each column corrupts one modality (Text / Audio / Visual) at increasing rates; rows correspond to datasets.

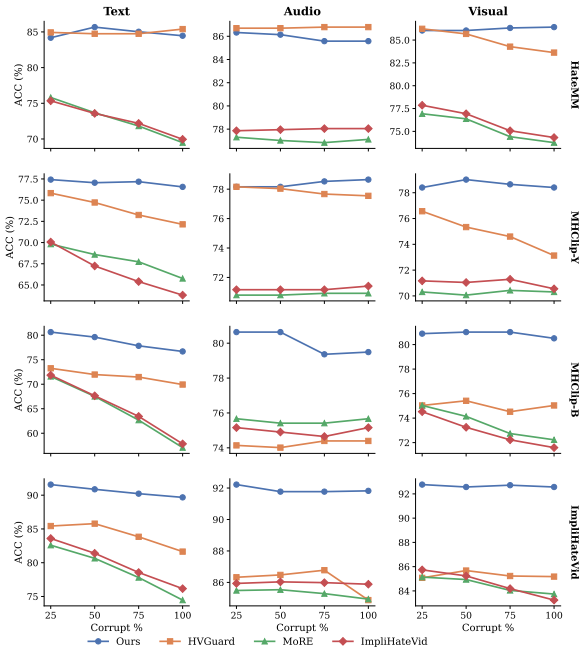


Figure 14: Modality robustness under test-time corruption.

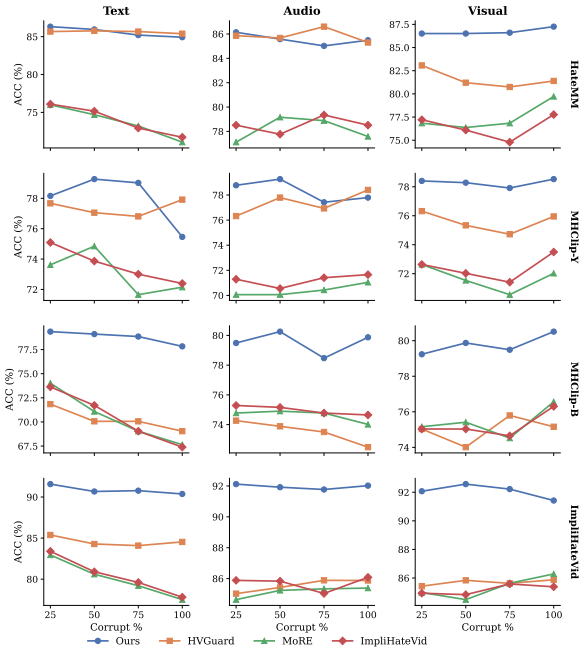


Figure 15: Modality robustness under train-and-test-time corruption.